Extensible Causality Does Not Guarantee Pure Quantum Dilations

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Every ordinary quantum channel admits a reversible explanation on a larger system. We show that extensible causality—causal records for every finite local experiment with arbitrary finite-dimensional shared inputs—does not guarantee such an explanation for a process connecting two laboratories. We construct a full-rank process on four qubit ports that satisfies this requirement yet has no pure process dilation with any finite environment. An exact operator identity, independent of ancillary dimension, establishes causality, while a causal witness and known pure-process structure exclude purification. The separation survives white-noise fractions below 1/11 and occupies a region of nonzero volume in valid process space. Entanglement-assisted causal tests are therefore insufficient to select processes admitting a reversible quantum dilation.

Introduction.—Ordinary quantum channels admit a reversible explanation on a larger system [1]. Allowing quantum control of laboratory order [2, 3] raises a physical consistency question: do causal experimental records guarantee that the connection retains such an explanation? Order control offers advantages in channel discrimination [4], computation [5], machine learning [6], and communication [7–9], with photonic demonstrations [10–13]. Higher-order transformations and quantum networks describe these connections [14–16], whereas process matrices do not assume a global causal order [17]. Purification extends the environmental explanation to these processes [18, 19]. Measurement reversibility [20] and information–disturbance trade-offs [21] illustrate why a classical record and the quantum information retained by an operation must be distinguished. For a connection between laboratories, we ask whether causal records ensure one reversible realization of the same process for all local operations.

Yet causal records need not reveal a process's realization. Causal inequalities constrain records [22]; causal witnesses test for a common quantum mixture of ordered processes [23, 24]. Feix et al.'s transpose construction [25] produces nonseparable processes with causal models without shared ancillas. With shared entanglement, transposing Bob's operations can also partially transpose the shared state and destroy positivity [25, 26]. Extensible causality requires causal records for every finite experiment with arbitrary finite-dimensional shared inputs [26]. A pure dilation must explain one process for all inserted operations [19]. Whether extensible causality guarantees separability or purification for two laboratories remained open [19, 26]. Pure-process structure [27, 28], spacetime realizations [29], and circuits for extensible causal separability [30–32] do not resolve this question.

Here we construct a full-rank process on four qubit ports that is causal for arbitrary finite shared inputs and all finite local experiments. Starting from the transpose construction, we derive a dimension-independent positive operator identity that supplies common order weights while allowing the branch records to depend on the experiment. A causal witness excludes any positive quantum mixture of orders, and known pure-process structure [27, 28] then rules out every finite pure dilation.

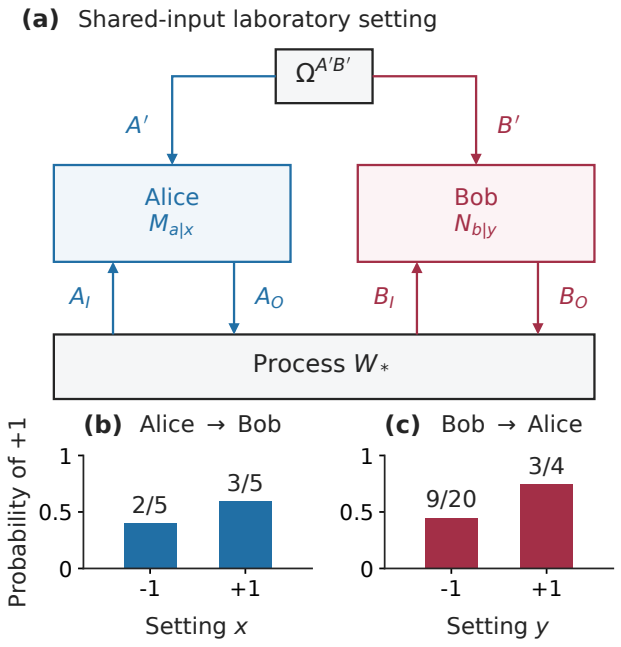


FIG. 1. Laboratory probes of W_*, the chosen counterexample. (a) Alice and Bob receive/return qubit ports marked I/O, share ancillary state $\Omega^{A'B'}$, and apply Choi operations $M_{a|x}, N_{b|y}$ with settings x, y and outcomes a, b. Links identify ports, not a realizing circuit. (b,c) Probabilities of outcome $+1$ for the two protocols in Eq. (4), with settings ± 1. Both directions respond; these experiments alone do not certify nonseparability.

Laboratory setting.—Atomic-ensemble communication [33], high-dimensional photonic-gate proposals [34], and teleportation-based quantum interfaces [35] use local quantum operations and shared resources. In the process description, these are specified separately from the connection between laboratories. Each laboratory receives a quantum input and returns a quantum output. Alice chooses a setting x, records a, and applies a completely positive instrument with Choi matrices $M_{a|x}$. Bob chooses y, records b, and uses $N_{b|y}$. The ports A_I, A_O, B_I, B_O belong to Alice and Bob, respectively; subscripts I, O denote input and output. With a shared input state $\Omega^{A'B'}$ on their local ancillary spaces A', B', the process matrix W gives

$$p(a, b|x, y) = \mathrm{Tr}[(W \otimes \Omega^{A'B'})(M_{a|x} \otimes N_{b|y})], \quad (1)$$

where the left side is the joint outcome probability conditional on the settings; Tr and $\otimes$ denote trace and tensor product. Summing each instrument over outcomes gives a trace-preserving channel. We use the full-transpose Choi convention in the Supplemental Material [36]. Process constraints ensure normalized, nonnegative probabilities without assuming a circuit [17].

A causal explanation of the records has the form

$$p = q\, p^{A \prec B} + (1 - q)\, p^{B \prec A}, \quad (2)$$

where p is the complete probability table, the superscripted terms are normalized ordered tables, and q is the setting-independent Alice-first probability. The notation $A \prec B$ means Alice's marginal is independent of y, while Bob may respond to x; $B \prec A$ reverses this condition. Extensible causality (EC) requires such a model for every finite shared ancillary state and every finite experiment, with all settings and outcomes [26]. The model may depend on the experiment. Causal separability (CS) requires one positive mixture of ordered *processes* reproducing all instruments.

Competing directions of influence.—All four original ports are qubits. Let $\sigma_x, \sigma_y, \sigma_z$ denote the Pauli matrices. A superscript identifies the port on which a matrix acts; omitted ports carry the identity. Products on distinct ports denote tensor products in the order A_I, A_O, B_I, B_O, and I is the identity on all four ports. Our example is

$$W_* = \frac{1}{4}\Big[I + \frac{1}{5}\sigma_z^{A_I} + \frac{3}{10}\sigma_z^{A_I}\sigma_x^{B_I}\sigma_z^{B_O} + \frac{1}{5}\sum_{k=x,y,z}\sigma_k^{A_O}\sigma_k^{B_I}\Big], \quad (3)$$

where k labels the three Pauli axes, and the asterisk labels this counterexample. It has trace four, obeys process normalization, and has minimum eigenvalue $1/40$, ensuring strict positivity. Full rank is not an assumption of the EC or nonpurifiability arguments below.

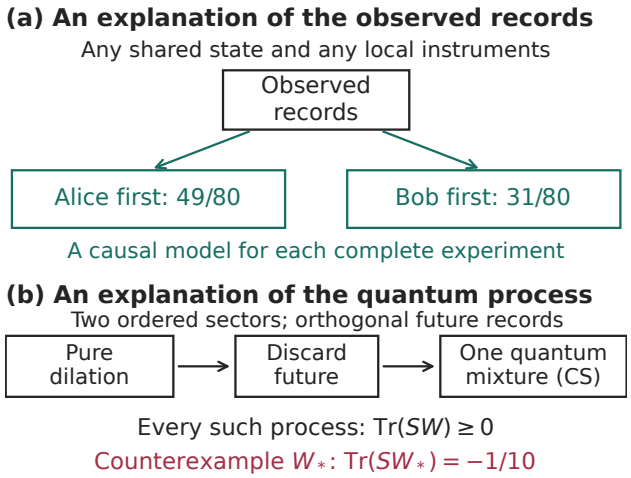


FIG. 2. Different demands on records and processes. (a) Every finite shared-input experiment on W_* admits Alice-first/Bob-first records with weights 49/80 and 31/80. Branch records may depend on the experiment. (b) In our bipartite setting, discarding the orthogonal future records of a pure dilation yields one positive mixture of ordered processes: causal separability (CS). The causal witness S has nonnegative trace pairing on this set, but is negative on the counterexample.

The term $\sigma_z^{A_I}$ biases Alice's input. The three A_O–B_I couplings transmit Alice's preparation to Bob's measurement; $\sigma_z^{A_I}\sigma_x^{B_I}\sigma_z^{B_O}$ carries reverse dependence through Alice's input and Bob's ports. Commutation with $\sigma_z^{A_I}$ and $\sigma_z^{B_O}$ makes these two ports classical labels in their respective bases. The other ports retain noncommuting Pauli correlations.

Two protocols expose these dependences (Fig. 1). First, Alice discards her input and prepares a $\sigma_x^{A_O}$ eigenstate with eigenvalue $x = \pm 1$. Bob measures $\sigma_x^{B_I}$, records $b = \pm 1$, and returns a maximally mixed output. Second, Bob measures $\sigma_x^{B_I}$, obtains $r = \pm 1$, and prepares a $\sigma_z^{B_O}$ eigenstate with eigenvalue yr; $y = \pm 1$ is his setting. Alice measures $\sigma_z^{A_I}$, records $s = \pm 1$, and returns a maximally mixed output. The Born rule gives

$$\langle b\rangle_x = \frac{x}{5}, \qquad \langle s\rangle_y = \frac{1}{5} + \frac{3y}{10}, \quad (4)$$

where angle brackets denote outcome averages conditional on the subscripted setting. Switching the settings changes the receivers' $+1$ probabilities by $1/5$ and $3/10$. A random mixture of orders can also display both responses in these separate experiments. The stronger question is whether *one quantum mixture* accounts for all operations on this process.

Why a pure dilation is excluded.—A pure dilation adds a global past and future, prepares a fixed pure past state, and discards the future [19]; their dimensions are arbitrary finite. Our original process has no global input or output. In this bipartite setting, a reversible process splits into two ordered sectors with orthogonal fu-

ture records [27, 28]. Discarding these records leaves a positive mixture of orders (Fig. 2). Conversely, standard ordered-comb dilation [16, 19] retains the environments of both branches and a record of the order to purify such a mixture. These established constructions give

$$\text{Purifiable} = \text{CS} \subseteq \text{EC}, \tag{5}$$

where the labels denote process sets. The argument allows arbitrary finite past and future dimensions, without a rank-based environment bound.

For W_*, a causal witness S made from a singlet projector on $A_O B_I$ and Pauli terms satisfies $\mathrm{Tr}(SW) \geq 0$ for every CS process, whereas

$$\mathrm{Tr}(SW_*) = -\frac{1}{10}. \tag{6}$$

The Supplemental Material proves witness positivity on both ordered cones [36]. Trusted process tomography reconstructs the Pauli coefficients of W from measured probabilities and knowledge of the calibrated local operations [37, 38]. The negative value excludes the common quantum mixture and hence every finite pure dilation. Quantum circuits with classical control of causal order (QC-CC) choose the next operation using classical records; circuits with quantum control (QC-QC) can keep this choice coherent [39]. For deterministic circuits using each operation once, dilating the internal maps and completing them to unitaries gives a pure extension. Preparing a fixed global input and discarding the entire future then yields a CS bipartite process, so these circuits cannot realize W_* in the present setting.

Causality survives arbitrary entangled probes.—Our result covers all local instruments, arbitrary finite shared entangled inputs, and all finite settings and outcomes. A single order weight $q = \frac{49}{80}$ works throughout; the causal explanation in Fig. 2(a) follows from an operator identity.

Reporting the two classical port labels refines outcomes; discarding them recovers the original probabilities. Each joint outcome's probability must be split between the two orders. We prescribe nonnegative first-party contributions $g_a(x)$ and $h_b(y)$ of total weights q and $1-q$, respectively. Each depends only on its own setting and remains unchanged across the other laboratory's choices.

The joint probabilities can be assigned to these branches precisely when every pair of accepted outcome groups $\mathcal{A}, \mathcal{B}$ obeys

$$p(\mathcal{A}, \mathcal{B}|x, y) \leq g(\mathcal{A}|x) + h(\mathcal{B}|y), \tag{7}$$

where both sides sum over the indicated outcome groups. Accepting both groups cannot exceed the prescribed first-party contributions. A capacitated transportation theorem converts these inequalities, including empty and full groups, into the complete causal model [36].

For a pure shared input with Bob marginal ρ, where $\mathrm{Tr}\,\rho = 1$, let τ_s be Bob's subnormalized conditional marginal for the accepted Alice group in input-flag sector $s = \pm 1$. Let E be Bob's accepted effect on $B_I B'$. Explicitly,

$$\begin{aligned} g(\mathcal{A}|x) &= \sum_{s=\pm 1} \frac{245 + 48s}{800} \mathrm{Tr}\,\tau_s, \\ h(\mathcal{B}|y) &= \frac{31}{160} \mathrm{Tr}[(I_{B_I} \otimes \rho)E], \\ g + h - p &= \frac{\langle \Psi|\mathcal{Q}|\Psi\rangle}{4\,000\,000} \geq 0, \end{aligned} \tag{8}$$

where I_{B_I} is the input-qubit identity, g, h, p in the last line abbreviate the three quantities in Eq. (7), and $|\Psi\rangle$ is the normalized shared input embedded in enlarged local ancillary spaces. The positive semidefinite operator $\mathcal{Q}$ is the sum of fifteen positive operator contractions printed in the Supplemental Material [36]. Its integer coefficient matrices have size at most four; the operators they weight may act on ancillary spaces of any finite dimension. The contributions have the required total weights q and $1-q$.

Retaining Alice's Kraus environment and applying a Naimark dilation to Bob's effects preserve the probabilities and these contributions for general instruments, including complex amplitudes and multiple Kraus branches. The identity uses only the positive remainder for Alice's rejected event, Bob's projector relations, and commutation between laboratories, so it holds in every finite ancillary dimension. Mixed shared states follow by convexity with the same q. No such experiment can therefore violate a causal inequality.

Stability under noise and general perturbations.—The separation has a finite noise tolerance. Let the uniform process $I/4$ replace W_* with probability η:

$$W_\eta = (1-\eta)W_* + \eta I/4, \qquad 0 \leq \eta \leq 1. \tag{9}$$

Replacement erases the directional responses with probability η. Convexity preserves EC along the whole line. The witness becomes $(11\eta - 1)/10$, excluding pure dilation below $1/11$, or approximately 9.1% white noise. An explicit positive ordered decomposition at the threshold [36] gives

$$\begin{aligned} &W_\eta \in \text{EC} \quad \forall \eta, \\ &W_\eta \in \text{Purifiable} \iff W_\eta \in \text{CS} \iff \eta \geq \frac{1}{11}, \end{aligned} \tag{10}$$

where the equivalent conditions identify a common noise threshold for causal separability and pure dilation. EC holds over the full interval [Fig. 3(a)].

General perturbations may also break the Pauli pattern. With $W_0 = I/4$ the uniform process, define the neighborhood center

$$\overline{W} = \frac{21}{22} W_* + \frac{1}{22} W_0. \tag{11}$$

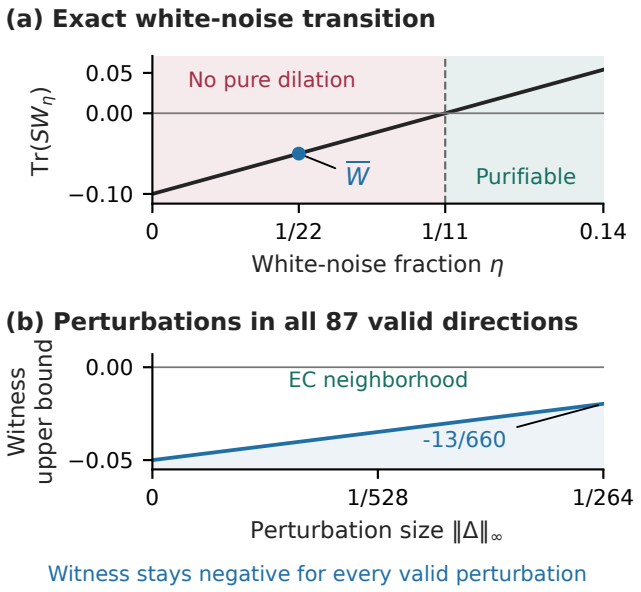


FIG. 3. Separation survives noise and general perturbations. (a) Exact trace pairing of witness S with the noisy process W_η; η is the uniform replacement probability. The whole noise line is EC. Shading meets at the exact pure-dilation threshold $1/11$; this plot enlarges its vicinity. For $\eta \geq 1/11$, the ordered branch weights are $3v/5$ and $1 - 3v/5$, where $v = 1 - \eta$; each weight is the component trace divided by four. The blue point is $\overline{W} = W_{1/22}$. (b) Upper bound $-1/20 + 8\|\Delta\|_\infty$ for that witness about $\overline{W}$; Δ is a valid Hermitian trace-zero perturbation and its norm is its largest absolute eigenvalue. Every such direction remains EC with negative witness within this sufficient radius, which is not an optimal boundary.

For every Hermitian, trace-zero perturbation Δ in the valid process linear subspace,

$$\|\Delta\|_\infty \leq \frac{1}{264} \implies \overline{W} + \Delta \in \mathrm{EC} \setminus \mathrm{CS}, \qquad (12)$$

where the norm is the largest absolute perturbation eigenvalue; the set difference means EC but not CS. For trace four, the bound covers all 87 valid directions, including broken classical-label symmetries. Mixing with a CS neighborhood of the uniform process retains EC and a strictly negative witness [36] [Fig. 3(b)]. The bound is sufficient and need not be optimal. It is expressed in operator norm rather than as an experimental error percentage, and it does not establish that W_* itself is an interior point of EC.

Discussion.—In our bipartite setting, Purifiable = CS follows from known pure-process structure and comb dilation [16, 19, 27, 28]. Establishing EC outside CS for arbitrary finite shared inputs answers the purification question in Ref. [19]. A causal model may adapt to each experiment; a pure dilation must explain one process across all local operations. For our process, trusted tomography uses calibrated operations to reconstruct this common operator; its negative witness value excludes every finite pure dilation. Thus causal-inequality compatibility alone cannot certify reversible realizability. The distinction is between a connection tested across all local operations and the properties of a specified dynamical model. Monitored scrambling studies information retention under measurements [40]. Atomic decay in a cavity [41] and entanglement loss [42] track the fate of excitations and correlations; dissipative excitation transfer [43] and effective non-Hermitian many-body models [44] address transport and collective phases. Our result concerns two laboratories with qubit ports and no original global past or future. The QC-QC consequence applies to standard deterministic circuits using each operation once [39]; postselected simulations, other realization frameworks, and multiple-use composition lie outside its scope.

In summary, compatibility with causal inequalities for every finite experiment with arbitrary finite-dimensional shared inputs does not guarantee the existence of a finite pure dilation. Reversible realizability is a constraint on the common quantum connection across all inserted operations.

Data availability.—The exact rational data and verification code are publicly available in the GitHub repository and in the accompanying supplemental files. They check the process, witness, laboratory responses, and stability without optimization.

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